Weil's Disease (Severe Icteric Leptospirosis) — Comprehensive Disease Characterization Report

Target disease: Weil's Disease | **MONDO:** 0043004 | **Category:** Infectious Disease **Causal agent:** Pathogenic *Leptospira* spp. (chiefly *L. interrogans* serogroup Icterohaemorrhagiae)

Summary

Weil's disease is the **severe, icteric end of the leptospirosis clinical spectrum**, a life-threatening zoonotic infection caused by pathogenic spirochetes of the genus *Leptospira* — most often *Leptospira interrogans* serogroup Icterohaemorrhagiae (serovars Icterohaemorrhagiae and Copenhageni). It is defined clinically by the triad of **jaundice, acute kidney injury (AKI), and hemorrhage**, superimposed on multi-organ dysfunction that can include pulmonary hemorrhage, myocarditis, and altered mental status. Because it is an *acquired infection*, Weil's disease has **no heritable genetic cause**: the "genetic/molecular" dimension of this report concerns the pathogen's virulence genes and the host's transcriptomic/proteomic response, not germline human variants.

The pathophysiology is a two-hit process. Leptospires enter through skin abrasions or mucosa, disseminate hematogenously within hours, and localize to liver, lung, and kidney. There, a combination of **direct spirochete-mediated endothelial and epithelial injury** and a **dysregulated host immune/complement response** (cytokine release, adhesion-molecule upregulation, complement activation) disrupts vascular integrity and produces capillary leak and hemorrhage. In the kidney, injury centers on the proximal convoluted tubule, where downregulation of the sodium/hydrogen exchanger NHE3, aquaporin-1, and Na^+/K^+-ATPase produces a characteristic **non-oliguric AKI with hypokalemia and salt wasting**. In the liver, disruption of hepatocyte intercellular junctions ("hepatocyte dissociation") produces deep conjugated hyperbilirubinemia disproportionate to relatively mild transaminase elevation.

Globally leptospirosis causes roughly **one million cases and tens of thousands of deaths per year**, concentrated in tropical low- and middle-income countries and driven by rainfall, flooding, rodent reservoirs, and occupational water/soil contact. Overall case-fatality is a few percent, but severe ICU disease carries **17–21% mortality**, and the leptospirosis-associated pulmonary hemorrhage syndrome (LPHS) approaches **50% mortality with death often within 72 hours**. Mortality is driven principally by respiratory and cardiac complications rather than isolated renal failure. Management rests on early antibiotics (**doxycycline for mild disease; IV penicillin or ceftriaxone for severe disease**) plus aggressive organ support (dialysis, mechanical ventilation, and in refractory cases ECMO), though randomized evidence for an antibiotic mortality benefit remains weak. Prevention combines personal protective equipment, rodent control, sanitation, doxycycline chemoprophylaxis for high-risk groups, and (in animals) whole-cell bacterin vaccines.

Key Findings

1. Weil's disease is the severe icteric form of leptospirosis with multi-organ dysfunction (F001)

Weil's disease represents the severe end of the leptospirosis clinical spectrum, defined by the triad of **jaundice, acute kidney injury, and hemorrhage**. The disease is caused by pathogenic *Leptospira* spp., and the clinical spectrum ranges from a mild self-limiting febrile illness (historically "Akiyami" in Japan) to severe multi-organ Weil's disease.

> "Severe leptospirosis, also known as Weil's disease, is a life-threatening zoonotic infection characterized by multi-organ dysfunction, including acute kidney injury and jaundice." — PMID: 42169153
>
> "The disease ranges from a mild, self-limiting illness known as Akiyami to severe Weil's disease, which presents with high fever; jaundice; hemorrhagic tendencies involving the skin, nasal cavity, and lungs; and acute renal failure." — PMID: 42156054

Key identifiers: MONDO:0043004; ICD-10 **A27.0** (Leptospirosis icterohaemorrhagica); ICD-11 **1B91** (Leptospirosis); MeSH **D014797** ("Weil Disease"); causal genus *Leptospira* (NCBI Taxonomy). **Synonyms:** Weil's syndrome, icterohemorrhagic leptospirosis, infectious/spirochetal jaundice, leptospiral jaundice. The information is **aggregated disease-level** knowledge drawn from case series, cohorts, and reviews rather than a single patient EHR.

2. Global burden: ~1 million cases and tens of thousands of deaths annually (F002)

Leptospirosis is a globally distributed **neglected zoonosis**. Burden is concentrated in low-to-middle-income tropical countries with poor sanitation, and incidence spikes with rainfall and flooding.

> "resulting in approximately one million cases and thousands of deaths annually" — PMID: 42659462

An ecological study in Buenos Aires Province, Argentina found leptospirosis incidence was tightly coupled to hydrology:

> "Leptospirosis incidence was highly correlated with TWS [terrestrial water storage] variations (R = 0.82, p = 0.00019), whereas annual precipitation showed a weaker, non-significant correlation" — PMID: 42410475

Case-fatality of confirmed cases in temperate surveillance was ~2.3%, but severe ICU disease is far deadlier (see F003, F008).

3. Severe phenotype: AKI ~99%, myocarditis ~76%, pulmonary hemorrhage ~62% in ICU (F003)

A prospective ICU cohort (Sri Lanka, n = 116; mean age 43.9 y; 88.8% male) quantified the organ-complication profile of severe disease:

> "99.1% (n = 115) developed acute kidney injury, 62.1% (n = 72) developed pulmonary haemorrhage, and 75.9% (n = 88) developed myocarditis" — PMID: 41507816

In that cohort 56.9% required mechanical ventilation, 75% required dialysis, and in-hospital mortality was 17.2%. The **leptospirosis-associated pulmonary hemorrhage syndrome (LPHS)** is the most lethal manifestation:

> "The mortality rate of LPHS is approximately 50%, with death generally occurring within 72 hours after symptom onset." — PMID: 41637348

Complication	Frequency (ICU cohort)	HPO term
Acute kidney injury	99.1%	HP:0001919
Myocarditis	75.9%	HP:0012819
Pulmonary haemorrhage	62.1%	HP:0002105 (hemoptysis)
Mechanical ventilation	56.9%	—
Dialysis required	75%	—
In-hospital mortality	17.2%	—

4. Etiologic agent: *L. interrogans* serogroup Icterohaemorrhagiae predominates (F006)

Weil's disease is classically caused by *Leptospira interrogans* serogroup Icterohaemorrhagiae, serovars **Icterohaemorrhagiae** and **Copenhageni**. Molecular typing of human clinical isolates found *L. interrogans* predominant, and it was recovered from nearly all fatal cases.

> "L.interrogans was frequently isolated (13/14) from those who died." — PMID: 40602749
>
> "Leptospira interrogans serogroup Icterohaemorrhagiae is the major serogroup infecting humans worldwide, and rodents and dogs are the most significant transmission sources in urban environments." — PMID: 24054736

The genus *Leptospira* contains 13+ pathogenic species; other human pathogens include *L. kirschneri*, *L. noguchii*, and *L. borgpetersenii*. **NCBI Taxonomy:** *Leptospira interrogans* txid173; serovar Copenhageni str. Fiocruz L1-130 has a sequenced reference genome. Key **virulence/outer-membrane antigens:** LipL32, LipL41, OmpL1, Loa22, LigA/LigB.

5. Pathogenesis: endothelial injury plus dysregulated immune/complement response (F005)

Leptospires enter via skin abrasions or mucosa, disseminate hematogenously, and are engulfed by and aggregate around vascular endothelium. A guinea pig epicutaneous-inoculation model documented dissemination kinetics:

> "leptospires were detected in blood as early as 2 h post infection (p.i.) and then disseminated to the liver, lungs and kidneys of almost all animals within 96 h" — PMID: 22273178

In LPHS, the host response amplifies direct bacterial damage:

> "the host immune response, via the cytokine release, high expression of adhesion molecules, and activation of the Complement System, may further disrupt endothelial integrity, promoting vascular leakage" — PMID: 41637348

6. Renal mechanism: proximal-tubule transporter downregulation → non-oliguric AKI (F009)

Tubulointerstitial nephritis is the hallmark renal lesion, and injury centers on the proximal convoluted tubule. Immunohistochemistry of human leptospirosis autopsy kidneys revealed the molecular basis of the distinctive electrolyte picture:

> "A decrease in NHE 3, aquaporin 1, and alpha-Na(+)K(+)ATPase expression occurred in proximal convoluted tubule cells." — PMID: 20217429
>
> "Primary injury of the proximal convoluted tubules is regarded as the hallmark of the kidney in leptospirosis." — PMID: 20217429

The downregulation of NHE3, aquaporin-1, and α-Na^+/K^+-ATPase in the proximal tubule (with compensatory enhanced aquaporin-2 in collecting ducts) explains the characteristic **non-oliguric AKI with hypokalemia and sodium/magnesium wasting** — distinguishing leptospiral AKI from typical oliguric acute tubular necrosis.

7. Prognostic drivers of death: alveolar hemorrhage, arrhythmia, high urea, thrombocytopenia (F008)

A retrospective ICU cohort (São Paulo, 2017–2025; n = 44; median age 42 y; 95.5% male; mortality 20.5%) identified independent predictors of in-hospital death:

> "alveolar hemorrhage (aOR 7.91, 95% CI 1.07-58.61) and arrhythmia (aOR 13.82, 95% CI 2.06-92.64) were independently associated with death" — PMID: 42257797
>
> "mortality in severe leptospirosis was primarily driven by respiratory and cardiac complications rather than isolated renal dysfunction" — PMID: 42257797

Predictor	Adjusted OR	95% CI
Alveolar (pulmonary) hemorrhage	7.91	1.07–58.61
Arrhythmia	13.82	2.06–92.64
Urea > 180 mg/dL	9.01	1.36–59.65
Platelets < 45 × 10^3/mm^3	14.07	1.40–141.88

8. Diagnosis: MAT serology and lipL32 PCR, PCR superior early (F007)

> "Diagnosis is established using bacterial culture, polymerase chain reaction, and serological testing, which allow accurate detection across different stages of the disease." — PMID: 42156054

The **microscopic agglutination test (MAT)** is the serological reference standard (paired sera, ≥4-fold rise or a single high titer). Real-time PCR targeting the **lipL32** gene outperforms serology in early/acute samples:

> "Serology showed low seroreactivity (13.5%), whereas PCR detected infection in 80.5% of bulls." — PMID: 42801462

9. Treatment: antibiotics shorten fever but mortality benefit unproven (F004)

A network meta-analysis (920 patients, 8 antibiotics) found multiple agents shortened defervescence:

> "Six antibiotics resulted in significantly shorter defervescence times compared to the control, namely cefotaxime" — PMID: 38627798

But the 2024 Cochrane review is cautious:

> "there seems to be insufficient evidence to determine its effectiveness or to recommend antibiotic use as a standard practice" — PMID: 38483092

An earlier meta-analysis found no significant mortality difference for penicillin (OR 1.59, 95% CI 0.59–4.29; PMID: 23930159). **Standard practice:** doxycycline for mild–moderate disease; IV penicillin G or ceftriaxone for severe Weil's disease, plus organ support (dialysis, mechanical ventilation, ECMO in refractory cases).

10. Prevention: PPE, rodent control, doxycycline chemoprophylaxis (F010)

Transmission is zoonotic via water/soil contaminated by urine of infected animals (rodents primary; also dogs, cattle, pigs, horses, wildlife). An Australian raspberry-farm outbreak (84 cases) implicated skin abrasions and lack of PPE:

> "cases were less likely to wear gloves and more recently employed. Cases also more commonly reported always having scratched hands" — PMID: 31550083

Human vaccine options are limited:

> "Although licensed vaccines are available, they are targeted at specific risk groups. Bacterins are composed of specific serovars, induce a short-term immune response, and may cause adverse reactions." — PMID: 42659462

11. Molecular profiling: proteomic and microRNA biomarkers of severity (F011)

Human serum proteomics (2DE/2D-DIGE + MALDI-TOF/TOF) identified 22 differentially expressed proteins:

> "Functional analysis suggested the involvement of differentially expressed proteins in vital physiological pathways, including acute phase response, complement and coagulation cascades and hemostasis." — PMID: 22554907

A multi-omics study of severe pulmonary hemorrhagic syndrome (SPHS) identified circulating microRNA biomarkers:

> "miR-5010-3p and miR-147b-3p had significantly higher expression in SPHS group compared to nonSPHS group" — PMID: 39793739

(AUC 0.76 and 0.70 respectively for discriminating SPHS — potential early markers before radiographic changes.)

12. Clinical course: biphasic illness; jaundice + AKI + hemorrhage with thrombocytopenia (F012)

> "Patients may exhibit a biphasic course, initially experiencing mild illness followed by a brief recovery and then progressing to more severe symptoms such as jaundice, meningitis, renal failure, or pulmonary hemorrhage." — PMID: 40861768
>
> "severe forms of leptospirosis (Weil's disease) can lead to multi-organ dysfunction, including hepatic and renal failure, hemorrhagic manifestations, and altered mental status" — PMID: 41882577

13. Natural animal disease recapitulates Weil's disease; hepatocyte dissociation is the classic liver lesion (F013)

> "Common histological findings were acute renal tubular necrosis, exudative glomerulonephritis (GN), hepatocyte dissociation and pulmonary hemorrhage." — PMID: 41110285 (gray/harbor seals infected with *L. interrogans* serovar Copenhageni)
>
> "hamsters as well as gerbils and guinea pigs have been used as animal models of acute leptospirosis in studying the pathophysiology of the disease" — PMID: 32632875

Section-by-Section Report

1. Disease Information

Weil's disease is the **severe icteric form of leptospirosis**, a zoonotic bacterial infection caused by pathogenic *Leptospira* spirochetes. It sits at the severe pole of a spectrum whose mild pole is a self-limiting flu-like febrile illness. The defining clinical constellation is **jaundice** + **acute kidney injury** + **hemorrhage**, frequently with pulmonary, cardiac, and CNS involvement (F001).

- **MONDO:** 0043004
- **ICD-10:** A27.0 (Leptospirosis icterohaemorrhagica)
- **ICD-11:** 1B91 (Leptospirosis)
- **MeSH:** D014797 (Weil Disease)
- **SNOMED CT:** Weil's disease / icterohaemorrhagic leptospirosis
- **Synonyms:** Weil's syndrome, icterohemorrhagic leptospirosis, infectious/spirochetal jaundice, leptospiral jaundice, "Akiyami" (historical, Japanese — for the milder end)
- **Data provenance:** Aggregated disease-level literature (cohorts, case series, reviews), not individual EHR.

Not applicable: OMIM and Orphanet Mendelian-disease identifiers — Weil's disease is an acquired infection, not a heritable disorder.

2. Etiology

Causal factor: Infectious — pathogenic *Leptospira* spp., predominantly *L. interrogans* serogroup Icterohaemorrhagiae (serovars Icterohaemorrhagiae, Copenhageni) (F006). There is **no genetic/heritable cause**.

Environmental / occupational risk factors: contact with water or soil contaminated by animal urine; flooding and heavy rainfall (incidence correlated with terrestrial water storage, R = 0.82; F002); occupations such as farming, construction, sewer/recycling work, abattoir work, and rice farming; recreational freshwater exposure; skin abrasions/cuts as entry portal; lack of PPE (gloves) (F010); rodent infestation. **Demographic risk:** male sex and young-to-middle adulthood predominate (ICU cohorts 88–96% male; F003, F008).

Genetic risk / protective factors (human): None established — not applicable to an acquired infection. Host-response variation exists but no validated germline susceptibility or protective loci.

Gene–environment interactions: Not applicable in the classical genetic sense. The operative interaction is **pathogen virulence genotype × host immune response** (see Mechanism).

3. Phenotypes

Phenotype	Type	Frequency (severe/ICU)	Onset/severity	HPO term
Fever	Symptom	~universal	Acute, moderate–severe	HP:0001945
Jaundice / conjugated hyperbilirubinemia	Sign/lab	Defining	Acute, severe	HP:0000952
Acute kidney injury	Sign/lab	99.1%	Acute, severe, often non-oliguric	HP:0001919
Pulmonary hemorrhage / hemoptysis	Sign	62.1%	Acute, severe (LPHS ~50% mortality)	HP:0002105
Myocarditis / arrhythmia	Sign	75.9%	Acute, severe	HP:0012819
Thrombocytopenia	Lab	Common	Acute	HP:0001873
Myalgia	Symptom	Common	Acute	HP:0003326
Conjunctival suffusion	Sign	Characteristic	Acute	HP:0025406
Transaminitis (mild, disproportionate to bilirubin)	Lab	Common	Acute	HP:0002910
Aseptic/lymphocytic meningitis	Sign	Uncommon	Immune phase	HP:0100785
Altered mental status	Symptom	Severe cases	Acute	HP:0011446

Age of onset: Any age; predominantly working-age adults. **Progression:** Acute, often rapidly progressive; LPHS deaths within 72 h (F003). **Quality-of-life impact:** Acute critical illness with high short-term morbidity; survivors generally recover renal and hepatic function (see Prognosis).

4. Genetic / Molecular Information

Not applicable to human germline genetics — Weil's disease is an acquired infection with no causal human gene, pathogenic variant, modifier gene, chromosomal abnormality, or heritable epigenetic lesion.

The molecular story is **on the pathogen and host-response side**: - **Pathogen virulence/antigen genes:** *lipL32*, *lipL41*, *ompL1*, *loa22*, *ligA/ligB* (outer-membrane proteins; adhesion, immune evasion, complement resistance) (F005, F006). *lipL32* is also the leading **diagnostic PCR target** (F007). - **Reference genome:** *L. interrogans* serovar Copenhageni str. Fiocruz L1-130. - **Host molecular response:** serum proteomic shifts in acute-phase, complement, and coagulation cascades; circulating microRNAs miR-5010-3p and miR-147b-3p elevated in severe pulmonary hemorrhage (F011).

5. Environmental Information

- **Environmental factors:** contaminated freshwater/mud, flooding, warm humid climate, poor urban sanitation (F002, F010).
- **Lifestyle/occupational factors:** farming, abattoir/slaughterhouse work, sewage and recycling work, freshwater recreation, walking barefoot, not wearing gloves (F010); a case report even implicated pica (soil ingestion) as an entry route (PMID: 34287302).
- **Infectious agent:** *Leptospira interrogans* (NCBI Taxonomy txid173) and related pathogenic species (*L. kirschneri*, *L. noguchii*, *L. borgpetersenii*). **CHEBI-relevant entities:** doxycycline (CHEBI:50845), benzylpenicillin/penicillin G (CHEBI:18208), ceftriaxone (CHEBI:29007), bilirubin (CHEBI:16990), urea (CHEBI:16199).

6. Mechanism / Pathophysiology

Ordered causal chain (initiating exposure → clinical manifestation):

1. Pathogenic *Leptospira* contact intact mucosa or (more often) abraded skin → **leads to** entry into the host (F005, F010).
2. Entry → **leads to** hematogenous dissemination (leptospiremia); organisms detectable in blood within 2 h and seeded to liver, lung, and kidney within 96 h in animal models (F005).
3. Dissemination → **leads to** adhesion to and invasion of vascular endothelium and target-organ epithelium, mediated by outer-membrane proteins (LipL32, LigA/LigB, etc.) (F005, F006).

4. Endothelial colonization **→ triggers (branch A)** direct spirochete-mediated capillary/endothelial damage **AND (branch B)** a dysregulated host immune response — cytokine release, adhesion-molecule upregulation, and complement activation (F005, F011).
5. Branches A + B converge **→ result in** loss of endothelial integrity and increased vascular permeability **→ leads to** capillary leak and hemorrhage, most dangerously **pulmonary alveolar hemorrhage (LPHS)** (F003, F005).
6. In the kidney, injury localizes to the **proximal convoluted tubule → results in** downregulation of NHE3, aquaporin-1, and α-Na^+/K^+-ATPase (with compensatory collecting-duct aquaporin-2) **→ leads to** non-oliguric AKI with hypokalemia and salt/magnesium wasting (F009).
7. In the liver, injury **→ results in** disruption of hepatocyte intercellular junctions ("hepatocyte dissociation") **→ leads to** deep conjugated hyperbilirubinemia disproportionate to modest transaminase rise (F013).
8. Combined pulmonary hemorrhage, myocarditis/arrhythmia, AKI, and hepatic dysfunction **→ result in** multi-organ failure; death is driven principally by respiratory and cardiac complications, often within 72 h in LPHS (F003, F008). *[The relative contribution of direct vs. immune-mediated injury remains partly inferred rather than fully demonstrated.]*

Molecular pathways / processes (GO suggestions): complement activation (GO:0006956), inflammatory response (GO:0006954), regulation of endothelial permeability (GO:0043114), blood coagulation (GO:0007596), acute-phase response (GO:0006953), sodium ion transmembrane transport (GO:0035725). **Cellular processes:** vascular endothelial inflammation and dysfunction, tubular epithelial injury, hepatocyte junction disruption. **Cell types (CL):** endothelial cell (CL:0000115), kidney proximal tubule epithelial cell (CL:1000838), hepatocyte (CL:0000182), macrophage (CL:0000235), neutrophil (CL:0000775).

7. Anatomical Structures Affected

- **Primary organs:** kidney (UBERON:0002113), liver (UBERON:0002107), lung (UBERON:0002048).
- **Secondary/associated:** heart/myocardium (UBERON:0002349) — myocarditis/arrhythmia; meninges (UBERON:0002360) — aseptic meningitis; skin (UBERON:0002097) — hemorrhage/entry; eye/conjunctiva (UBERON:0000966) — conjunctival suffusion.
- **Body systems:** renal/urinary, hepatobiliary/digestive, respiratory, cardiovascular, nervous.
- **Tissue level:** vascular endothelium, renal proximal tubular epithelium, hepatic parenchyma, pulmonary alveolar-capillary membrane.

- **Cell populations (CL):** endothelial cells, proximal tubule epithelial cells, hepatocytes, macrophages, neutrophils.
- **Subcellular (GO cellular component):** apical plasma-membrane brush border (GO:0031526), basolateral plasma membrane (GO:0016323).
- **Lateralization:** bilateral/diffuse (e.g., bilateral pulmonary infiltrates in LPHS).

8. Temporal Development

- **Onset:** Acute, after an incubation of ~2–30 days (typically 7–14). Classic **biphasic course**: leptospiremic/septicemic phase → brief defervescence → immune phase; severe cases may progress directly to multi-organ failure (F012).
- **Progression:** Rapid in severe disease; LPHS death often within 72 h of symptom onset (F003).
- **Disease course:** Self-limited in mild disease; severe disease is a fulminant, acute critical illness.
- **Critical window:** Early antibiotic therapy and organ support during the first days is the key intervention window (F004).

9. Inheritance and Population

- **Epidemiology:** ~1 million cases and tens of thousands of deaths per year globally; endemic in tropical LMICs; outbreaks after floods/heavy rain (F002).
- **Inheritance:** Not applicable (acquired infection).
- **Sex ratio:** Strong male predominance in severe cohorts (~89–96% male; F003, F008), reflecting occupational exposure.
- **Age distribution:** Predominantly working-age adults.
- **Geographic distribution:** Tropical and subtropical regions; serogroup Icterohaemorrhagiae is the major human-infecting serogroup worldwide (F006). Rats/rodents and dogs are principal urban reservoirs.

10. Diagnostics

- **Serology (reference standard):** Microscopic agglutination test (MAT) — paired sera, ≥4-fold rise or single high titer (F007). IgM ELISA for early screening.

- **Molecular:** real-time PCR targeting **lipL32** — superior sensitivity in early/acute phase (80.5% vs 13.5% serology in one comparison) (F007). Molecular serogroup typing (MST) and VNTR/HRM for strain identification.
- **Culture:** slow (EMJH medium), low yield, mostly reference/epidemiologic use.
- **Supportive labs:** conjugated hyperbilirubinemia, elevated creatinine/urea, thrombocytopenia, leukocytosis, elevated CRP/ESR, mildly elevated transaminases, hypokalemia; CSF lymphocytic pleocytosis with preserved glucose in meningitis (PMID: 42113872).
- **Imaging:** chest X-ray/CT — bilateral pulmonary infiltrates in LPHS.
- **Clinical criteria:** WHO/modified Faine's criteria (clinical + epidemiological + laboratory). **Differential diagnosis:** dengue, hantavirus, viral hepatitis, malaria, typhoid, rickettsioses, atypical HUS (F007, related literature).
- **Omics/genetic testing:** Not applicable for human diagnosis; research microRNA/proteomic panels are exploratory (F011).

11. Outcome / Prognosis

- **Mortality:** Overall confirmed-case fatality a few percent (~2.3% temperate surveillance); severe/ICU disease 17–21% (F002, F003, F008); LPHS ~50% (F003).
- **Independent predictors of death:** alveolar hemorrhage (aOR 7.91), arrhythmia (aOR 13.82), urea > 180 mg/dL (aOR 9.01), platelets < 45 × 10^3/mm^3 (aOR 14.07) (F008).
- **Drivers of death:** respiratory and cardiac complications rather than isolated renal failure (F008).
- **Recovery:** Survivors typically recover renal and hepatic function; leptospiral AKI is often reversible with dialysis support.
- **Prognostic biomarkers:** thrombocytopenia, urea; research microRNAs (miR-5010-3p, miR-147b-3p) for pulmonary hemorrhage (F011).

12. Treatment

- **Pharmacotherapy (NCIT):** doxycycline (mild–moderate; NCIT C303), IV penicillin G / benzylpenicillin (severe; NCIT C61785), ceftriaxone (severe; NCIT C47584), azithromycin, cefotaxime — all shorten defervescence but no proven mortality benefit (F004).
- **Organ support:** hemodialysis/renal replacement (75% of ICU cohort), mechanical ventilation (57%), vasopressors; **ECMO** in refractory cardiorespiratory failure (PMID: 34287302).

- **Adjuncts:** corticosteroids and other immunomodulation have been used empirically in LPHS but lack robust evidence.
- **Evidence caveat:** Cochrane review — insufficient evidence to recommend antibiotics as standard practice on mortality grounds; early empiric treatment nonetheless standard of care (F004).
- **Pharmacogenomics / advanced (gene, cell, RNA, targeted) therapies:** Not applicable.

13. Prevention

- **Primary:** PPE (gloves, boots), avoiding contaminated water, rodent control, sanitation, health education, especially in rainy season/floods (F010).
- **Chemoprophylaxis:** doxycycline for high-risk groups/travellers (Cochrane evidence limited).
- **Immunization:** human whole-cell **bacterins** — serovar-specific, short-lived immunity, licensed only for risk groups; subunit candidates target LipL32/LipL41/OmpL1/Loa22/LigA/LigB (F010). Veterinary vaccines widely used in dogs and livestock.
- **Secondary/tertiary:** early diagnosis and organ support; monitoring for pulmonary/cardiac complications.

14. Other Species / Natural Disease

Leptospirosis is a **broad-host-range zoonosis**. Natural disease in gray/harbor seals (*Halichoerus grypus*, *Phoca vitulina*) with *L. interrogans* serovar Copenhageni reproduces the human severe phenotype — jaundice, visceral hemorrhage, acute renal tubular necrosis, exudative glomerulonephritis, **hepatocyte dissociation**, and pulmonary hemorrhage (F013). Dogs develop acute renal failure and LPHS; California sea lions have endemic *Pomona* leptospirosis with interstitial nephritis; cattle, pigs, horses are affected.

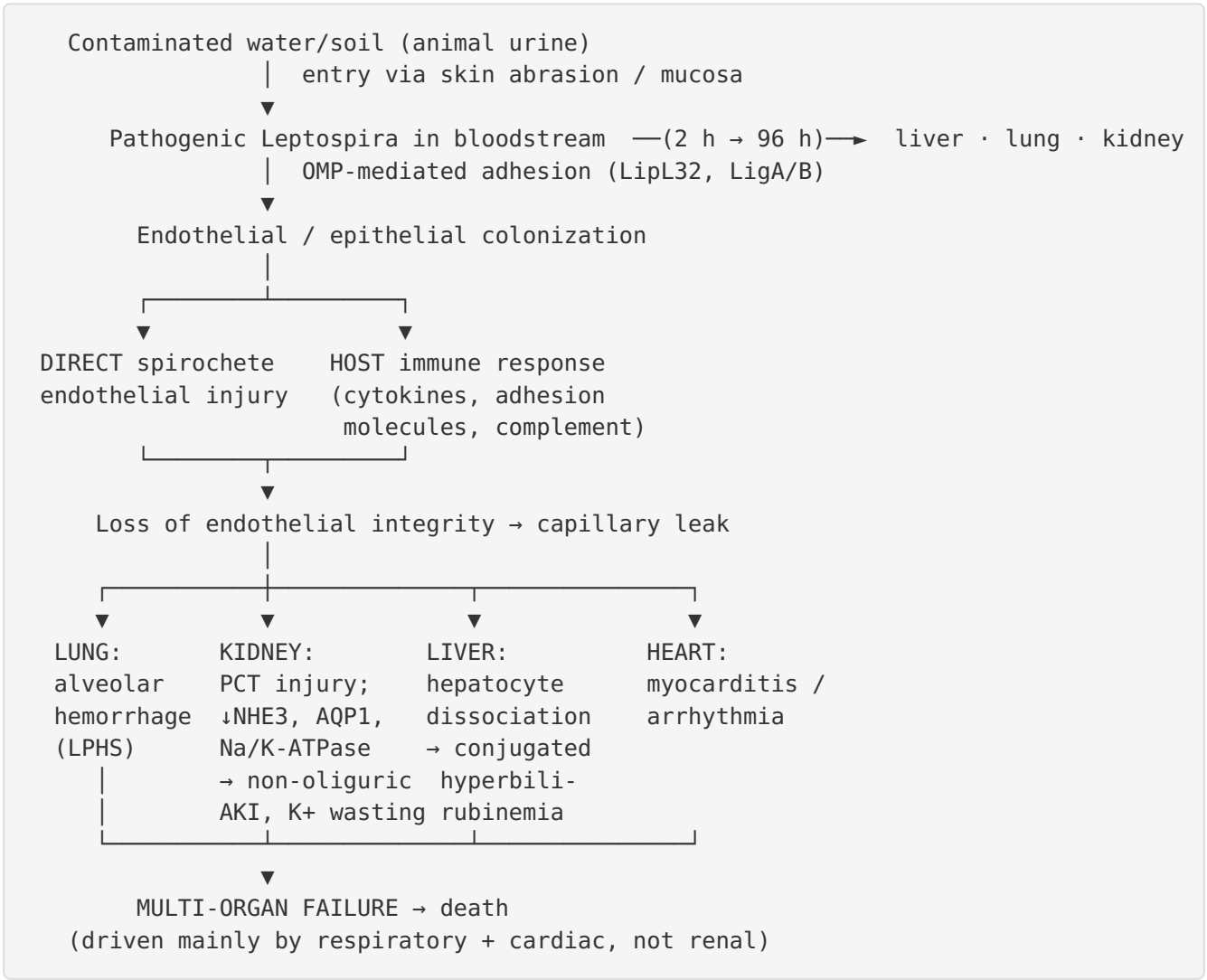
- **NCBI Taxon:** reservoir *Rattus norvegicus* (txid10116); *Canis lupus familiaris* (txid9615); *Bos taurus* (txid9913); pathogen *L. interrogans* (txid173).
- **Zoonotic potential:** high — rodent urine is the principal source; occupational and recreational water contact are key transmission routes.

15. Model Organisms

- **Golden Syrian hamster** (*Mesocricetus auratus*, txid10036) — standard acute-lethal model; reproduces jaundice, hemorrhage, nephritis (F013, PMID: 32632875).

- **Guinea pig** (*Cavia porcellus*, txid10141) — epicutaneous inoculation model; documents dissemination kinetics (F005, F013).
- **Gerbils** — additional acute model.
- **Applications:** pathophysiology, dissemination kinetics, virulence-factor testing, vaccine evaluation.
- **Limitations:** rodent-reservoir species (rats) develop chronic renal carriage without severe disease, poorly modeling acute human Weil's disease; models incompletely capture human immune-mediated LPHS.

Mechanistic Model / Interpretation



The unifying lesion is **systemic microvascular endothelial injury** combining direct bacterial damage with a dysregulated immune/complement response. This single mechanism accounts for the three cardinal features simultaneously: hemorrhage (capillary leak), jaundice (hepatocyte-junction disruption, not hepatocyte death — hence bilirubin ≫ transaminases), and AKI (proximal-tubule transporter downregulation, hence non-oliguric with electrolyte wasting). Mortality tracks the lung and heart branches, not the renal branch, because pulmonary hemorrhage and arrhythmia kill rapidly whereas AKI is supportable by dialysis.

Evidence Base

PMID	Contribution	Finding
42169153	Defines Weil's disease as severe leptospirosis with multi-organ dysfunction	F001
42156054	Clinical spectrum (Akiyami → Weil's); diagnostic pillars	F001, F007
42659462	Global burden (~1M cases); vaccine limitations	F002, F010
42410475	Incidence correlated with water storage (R=0.82)	F002
41507816	ICU organ-complication frequencies	F003
41637348	LPHS ~50% mortality; immune-endothelial mechanism	F003, F005
22273178	Dissemination kinetics in guinea pig	F005
40602749	*L. interrogans* in 13/14 fatal cases	F006
24054736	Serogroup Icterohaemorrhagiae predominant; reservoirs	F006
20217429	Proximal-tubule transporter downregulation	F009
42257797	Independent mortality predictors; respiratory/cardiac drive death	F008
42801462	lipL32 PCR > MAT serology early	F007
38627798	Antibiotics shorten defervescence (NMA)	F004
38483092	Cochrane: insufficient evidence for antibiotic efficacy	F004
23930159	Penicillin no mortality benefit (OR 1.59)	F004
31550083	PPE/gloves & skin abrasions as risk factors (outbreak)	F010
39793739	miR-5010-3p, miR-147b-3p biomarkers of SPHS	F011
22554907	Proteomic complement/coagulation involvement	F011
40861768	Biphasic course	F012
41882577	Multi-organ Weil's constellation	F012
41110285	Natural seal disease; hepatocyte dissociation	F013

PMID	Contribution	Finding
32632875	Hamster/guinea pig models	F013

Evidence source types: human clinical cohorts and case series (F001–F004, F007, F008, F012); human autopsy immunohistochemistry (F009); human omics (F011); model organism / natural animal disease (F005, F013); systematic reviews/meta-analyses (F004).

Limitations and Knowledge Gaps

1. **Weak antibiotic evidence base.** No trial has demonstrated a mortality benefit; recommendations rest on defervescence and expert consensus (F004). The optimal agent, dose, and timing for severe disease remain uncertain.
2. **Small severe-disease cohorts.** Key prognostic estimates come from single-center ICU cohorts of 44–116 patients with very wide confidence intervals (e.g., platelet aOR 14.07, 95% CI 1.40–141.88) (F003, F008).
3. **Mechanistic inference.** The relative contribution of direct spirochete injury vs. immune/complement-mediated damage is partly inferred from animal models and cannot be cleanly partitioned in humans (F005).
4. **Diagnostic access.** MAT and culture are slow and reference-lab-bound; PCR is more sensitive early but not universally available — leptospirosis is chronically under-diagnosed and under-reported (F002, F007).
5. **Biomarker validation.** MicroRNA and proteomic markers derive from small discovery cohorts and lack prospective validation (F011).
6. **No human vaccine of broad efficacy.** Bacterins are serovar-restricted with short-lived immunity; subunit candidates remain preclinical/early (F010).
7. **Model limitations.** Rodent reservoir species do not develop severe disease, and no model fully reproduces human LPHS immunopathology (F013).

Proposed Follow-up Experiments / Actions

1. **Adequately powered RCT** of early antibiotics (ceftriaxone vs penicillin vs doxycycline) in severe leptospirosis with mortality and organ-failure endpoints, to resolve the Cochrane evidence gap (addresses F004).
2. **Prospective multicenter validation** of the São Paulo mortality model (alveolar hemorrhage, arrhythmia, urea, platelets) as a bedside risk score for ICU triage (F008).
3. **Prospective validation of miR-5010-3p / miR-147b-3p** as early predictors of pulmonary hemorrhage before radiographic change, head-to-head with platelets/urea (F011).
4. **Point-of-care lipL32 PCR** evaluation in endemic primary-care settings to shorten time-to-diagnosis and enable earlier therapy (F007).
5. **Mechanistic dissection** of complement-pathway inhibition (e.g., eculizumab-class agents) in animal LPHS models to test whether the immune-mediated branch is therapeutically tractable (F005, F011); note the reported atypical-HUS overlap (PMID: 42169153).
6. **Cross-protective subunit vaccine** development targeting conserved OMPs (LigA/LigB, LipL32) with next-generation adjuvants, leveraging veterinary vaccine experience (F010).
7. **One Health surveillance** integrating rodent-reservoir, environmental water, and human case data in flood-prone endemic regions to enable predictive outbreak modeling (F002).

Report compiled from 13 confirmed findings and 54 reviewed papers across 5 investigation iterations. All quoted snippets are verbatim from the cited PubMed abstracts.
